PatientTriage.ai

“Triage is a snapshot. Risk isn't.”

A Continuous Safety Decision-Support Layer for Emergency Waiting Rooms • Accenture Innovation Challenge 2026

COMPETITION TRACK	CORE STACK	DEPLOYMENT	REPOSITORY
Round 2 Technical Prototype	Python 3.13 / FastAPI / React 19	Edge / On-Premise Air-Gapped	github.com/freya1705/PatientTriage---Accenture-

1. Project Overview & Introduction

PatientTriage.ai is an emergency department clinical decision-support system and continuous physiological safety surveillance layer. While traditional emergency triage treats patient prioritization as a static, one-time snapshot at hospital intake, PatientTriage.ai continuously tracks patient waiting times, vital sign trajectory velocity (ΔSpO■, ΔHR), data uncertainty, and active clinical attention.

The platform answers the single most critical operational question facing emergency clinicians: ***“Who in the waiting room is no longer safe to keep waiting?”***

2. The Core Problem: 3 Failure Modes of Traditional Triage

In emergency departments, triage is usually a point-in-time snapshot. Once triaged, patients wait for hours where risk continuously changes. Traditional static triage exhibits 3 concrete failure modes:

- **1. Silent Post-Triage Waiting Room Deterioration:** Patients triaged as Level 3 (Urgent) or Level 4 (Less Urgent) silently deteriorate in the waiting lounge without triggering an alert until physical collapse.
- **2. Missing Vitals & Stale Data Assumed Safe:** Incomplete records lacking SpO■ or BP default to low acuity, creating false reassurance. In medicine, *Unknown is NOT Safe*.
- **3. The Clinical Attention Bottleneck:** Static lists rank patients purely by intake severity. Attended critical patients block the queue, while *unattended deteriorating patients* remain hidden.

3. System Architecture: 3-Tier Layered Design

To ensure patient safety, mathematical explainability, and ethical clinician governance, PatientTriage.ai implements a strict 3-tier separation:

Tier	Responsibilities & Scope	Key Modules
Tier 1: Deterministic Safety Layer	Deterministic red-flags (SpO■ < 85%, SBP < 75 mmHg, FAST Stroke, pediatric stridor) that bypass ML models. Counterfactual downgrade safety blocking.	`safety_guardrails.py` `downgrade_guard.py`
Tier 2: AI & Decision Support	Continuous vital trajectory velocity (ΔSpO■, ΔHR), dynamic confidence decay ($\tau_{staleness}$), uncertainty scoring (Unknown ≠ Safe), Attention Gap re-ranking.	`risk_engine.py` `deterioration_engine.py` `attention_gap_engine.py`
Tier 3: Clinician Governance	Licensed clinician override authority, mandatory justification recording, and immutable append-only audit logging.	`audit_service.py` `OverrideModal.jsx`

4. Core Intelligence Engines & Mathematical Formulations

A. Age-Aware Physiological Calibrations: Pediatric models calibrate for infant tachycardia (>160 bpm) and toddler high fever (≥38.5°C in <3yo) with minimum blood pressure formula $SBP_{min} = 70 + (2 \times Age)$. Geriatric models calibrate for hypothermic occult sepsis (<36.0°C) and elevated shock thresholds (SBP < 100 mmHg).

B. Uncertainty-as-Risk Engine (“Unknown ≠ Safe”): Missing critical vitals deduct confidence (Σ SpO■: +22%, SBP: +20%, Zero History: +18%). Asymmetric safety bias escalates low-acuity cases with missing vitals to Level 3 Urgent and mandates an [ACQUIRE VITALS] verification.

C. Dynamic Confidence Decay & Safety Expiry ($\tau_{staleness}$): Evidence decays over time via $Confidence(t) = Base \times \max(0.20, 1.0 - [(t - t_{last}) / (Window \times 1.5)] \times 0.65)$. Exceeding the window flips status to SAFETY_EXPIRED.

D. The Attention Gap Priority Equation:

```
Action Priority Score = (w_r × Risk + Urgency) + (w_d × Deterioration) + (w_s × Staleness) + Wait Hazard + (w_u × Uncertainty) - (w_c × Clinical Coverage)
```

Where w_c deducts priority when an attending physician is already actively managing the patient (is_attended = True), allowing unattended deteriorating waiting patients to surface to Rank #1.

5. Benchmark Cohort & Empirical Impact Evaluation

Evaluated across 20 synthetic clinical scenarios representing 5 systematic emergency department failure modes:

Performance Dimension	Traditional Static Triage	PatientTriage.ai	Impact Delta
Waiting Deterioration Catch Rate	0% (Undetected)	**100% (Continuous)**	**+100% Safety Catch**
Stale Observation Flagging	0% (Assumed Safe)	**100% (Flagged EXPIRED)**	**Zero Unmonitored Stale Waits**
False Reassurance on Missing Vitals	High (Treated Normal)	**0% (Unknown ≠ Safe)**	**Eliminates Under-Triage**
Attention Gap Optimization	None (Attended Block)	**Active (Elevates Unattended)**	**Optimized Clinician Utilization**
Unsafe Priority Downgrades Blocked	0 Guardrails	**100% Guarded**	**100% Downgrade Guarded**

**Note: Results reflect simulated evaluations across 20 synthetic clinical benchmark scenarios for prototype demonstration.*

6. Key Features & Clinical Command Center UX

- **3-Zone Clinical Cockpit:** Left navigation rail, Center Live Action Queue workspace, and Right persistent safety summary panel.
- **Hero Live Action Queue:** Real-time rank ordering with SpO■ sparklines, Attention Gap meters, and one dominant Next-Best-Action button.
- **7 One-Click Demo Presets:** Instant simulation of toddler fever, geriatric sepsis, missing vitals, ambiguous diabetic nausea, and zero-history trauma.
- ■ **3× Surge Mode Disaster Simulator:** One-click volume expansion (20 → 60 patients) with automatic queue compression to prevent nurse alarm fatigue.
- **Hospital Profile Switcher:** Configurable between Urban Level-1 Trauma Center and Community Rural Clinic models.

7. Installation, Quick Start & Automated Verification

```
# 1. One-Command Launch (Windows PowerShell):
.\start.ps1

# 2. Manual Startup:
pip install -r requirements.txt && python -m uvicorn backend.main:app --reload --port 8000
cd frontend && npm install && npm run dev

# 3. Run Automated Test Suite (33 Unit & Integration Tests):
python -m pytest -v
```

8. Security, Privacy-by-Design & Governance

- **Zero PHI:** 100% synthetic physiological datasets without any real patient records.
- **Air-Gapped & On-Premise:** Runs entirely within hospital local network with zero external cloud LLM API dependencies.
- **Append-Only Audit Ledger:** Every intake, vital update, deterioration event, and clinician override is recorded with timestamps and clinical roles.

Regulatory & Safety Notice: PatientTriage.ai is a clinical decision-support research prototype developed for the Accenture Innovation Challenge 2026. All patient cohorts are synthetically generated. This system is not a certified medical device and does not replace licensed clinical judgment.